

EXPERIMENT-20

Implement Future Sales Prediction using a suitable Machine Learning algorithm.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
print("=" * 70)
print("      FUTURE SALES PREDICTION")
print("=" * 70)
uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("\nSALES DATASET")
print("-" * 70)
print(data.head(10))
print("\nColumns:")
print(list(data.columns))

# Convert categorical columns
for col in data.columns:
    if data[col].dtype == "object":
        data[col] = pd.factorize(data[col])[0]

# Convert date if available
for col in data.columns:
```

```

if "date" in col.lower():
    try:
        data[col] = pd.to_datetime(data[col])
        data[col] = (
            data[col].dt.year * 10000 +
            data[col].dt.month * 100 +
            data[col].dt.day
        )
    except:
        pass

# Find sales column
target = None
for col in data.columns:
    if "sales" in col.lower():
        target = col
        break

if target is None:
    target = data.columns[-1]
print("\nTarget Column :", target)

# Use numerical columns
numeric = data.select_dtypes(include="number")
X = numeric.drop(columns=[target])
y = numeric[target]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
model = RandomForestRegressor(
    n_estimators=100,
    random_state=42

```

```

)

print("\nTraining Future Sales Prediction Model...")

model.fit(X_train, y_train)

prediction = model.predict(X_test)

print("\nTraining Completed Successfully.")

print("\nActual Sales:")
print([round(x, 2) for x in list(y_test[:15])])

print("\nPredicted Sales:")
print([round(x, 2) for x in list(prediction[:15])])

print("\nR2 Score:")
print(round(r2_score(y_test, prediction), 3))
print("\nMean Squared Error:")
print(round(mean_squared_error(y_test, prediction), 3))
print("\n" + "=" * 70)
print("FUTURE SALES PREDICTION COMPLETED")
print("=" * 70)

```

Sample Input:

```

... =====
                        FUTURE SALES PREDICTION
=====
Choose Files sales data file.csv
sales data file.csv(text/csv) - 4062 bytes, last modified: 8/10/2026 - 100% done
Saving sales data file.csv to sales data file.csv

```

Sample Output:

```
SALES DATASET
...
    TV  Radio  Newspaper  Sales
0  230.1   37.8     69.2   22.1
1   44.5   39.3     45.1   10.4
2   17.2   45.9     69.3   12.0
3  151.5   41.3     58.5   16.5
4  180.8   10.8     58.4   17.9
5    8.7   48.9     75.0    7.2
6   57.5   32.8     23.5   11.8
7  120.2   19.6     11.6   13.2
8    8.6    2.1      1.0    4.8
9  199.8    2.6     21.2   15.6

Columns:
['TV', 'Radio', 'Newspaper', 'Sales']

Target Column : Sales

Training Future Sales Prediction Model...

Training Completed Successfully.

Actual Sales:
[16.9, 22.4, 21.4, 7.3, 24.7, 12.6, 22.3, 8.4, 16.5, 16.1, 11.0, 8.7, 16.9, 5.3, 10.3]

Predicted Sales:
[np.float64(17.3), np.float64(22.48), np.float64(19.77), np.float64(6.78), np.float64(22.96), np.float64(13.47), np.float64(22.59), np.float64(9.58), np.float64(16.89), np.fl

R2 Score:
0.953

Mean Squared Error:
1.437

=====
FUTURE SALES PREDICTION COMPLETED
=====
```